MOOC Dropout Prediction using Behavioural Sequences and Deep Learning

Mid Evaluation

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Research Domain: Educational Data Mining / Adaptive Learning Systems

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MOOC Dropout Problem

- MOOCs offer free and accessible online courses
- Major issue: very high student dropout rates
- Only about 13% of learners complete the courses
- Nearly 80–85% students drop out before finishing

Research Area

- MOOC dropout prediction
- Earlier dropout prediction

Identified Gaps

- Sequential data is available, but using multiple features in a sequence (multivariate) is limited
- Incorporation of learning style information is limited
- Early dropout prediction is limited

Limitations of Existing Models

- Existing dropout prediction models mostly use aggregated temporal features (e.g., weekly video plays, quiz attempts)
- Sequential patterns in student activity are underutilized
- Learning style information is rarely used to improve dropout prediction

Objectives

- Overcome the above issues

Dataset Overview

- We analyzed the KDDCup 2015 dataset
- Dataset contains folders:
  - train
  - test
  - date.csv
  - object.csv

Train Folder Files

- The train folder contains:
  - log_train
  - enrollment_train
  - truth_train

log_train Format

- Columns: enrollment_id, time, source, event, object
- enrollment_id: Unique ID for each student–course enrollment
- One student can have multiple enrollment_ids if they join multiple courses
- time format: 2014-06-14T09:38:29
- source types: server / browser
- event types: navigate, access, problem, page_close, video, discussion, wiki
- object: ID of the learning content

Sample Log Events

- Example records:

enrollment_id	time	source	event
1	2014-06-14T09:38:29	server	navigate
1	2014-06-14T09:38:39	server	access

enrollment_train Format

- Columns: enrollment_id, username, course_id
- enrollment_id: Unique ID for each student–course enrollment
- username: Unique ID for the student
- Same username appears if the student enrolls in many courses
- course_id: Course in which the student enrolled

truth_train Format

- Columns: enrollment_id, dropout
- dropout values:
  - 0 = No dropout
  - 1 = Dropout

date.csv Format

- Columns: course_id, from, to
- Contains course start and end dates

Example:

course_id	from	to
bWdj2...	2014-05-26	2014-06-24
RXDvf...	2014-05-25	2014-06-23

object.csv Format

- Columns: course_id, module_id, category, children, start
- course_id: Course identifier
- module_id: Unique ID of a course module
- category types include: about, chapter, course, html, problem, sequential, video, discussion, etc.
- children: IDs of sub-modules
- start: Date when the module becomes available

Why object.csv is Useful

- The object column in log_train refers to module_id in object.csv
- Helps identify the type of learning content used by the student

Test Folder Files

- The test folder contains:
  - log_test
  - enrollment_test
  - truth_test

Dataset Shapes

- log_train: (8157277, 5)
- object: (27249, 5)
- date: (39, 3)
- train_truth: (120541, 2)

Data Mapping

- log_train columns: enrollment_id, source, event, object
- object.csv columns: module_id, category
- object maps to module_id

Token Generation

- After mapping:
- enrollment_id, source, event, category
- Token format:

$$token = source + event + category$$

Examples:

- server_navigate_chapter
- browser_problem_problem
- browser_video_video

Temporal Feature

- Another evaluated item: time gap between events
- Time gap represents the interval between two consecutive actions

Current Implementation

The current system predicts MOOC dropout using event-level behavioral sequences extracted from interaction logs.

- Student actions are modeled as ordered sequences of learning events with temporal gaps, capturing patterns of engagement over time.
- These sequences are processed using a Bidirectional Long Short-Term Memory (BiLSTM) network to learn behavioral patterns associated with dropout.
- The model outputs the probability of dropout for each student based on their interaction sequence.

Event-Level Behavioral Representation

Key Idea

- Instead of aggregated activity counts, we model fine-grained interaction sequences of students.
- Each interaction event is converted into a behavioral token.

Event Token Construction

$$token = source + "_" + event + "_" + category$$

Examples

- browser_video_video
- browser_problem_problem
- server_access_sequential

Temporal Feature

$$gap = \log(1 + seconds_since_previous_event)$$

Behavioral Sequence Construction

Event Ordering

- Events are grouped by `enrollment_id`.
- Sorted by timestamp to maintain chronological order.

Token Sequence Generation

Example sequence:

- browser_video_video
- browser_problem_problem
- server_access_sequential
- browser_video_video

Add Temporal Gaps

Each event is paired with its time gap:

$$(event_token, time_gap)$$

Sequence Length Standardization

- Maximum sequence length: 150 events
- Short sequences are padded
- Long sequences are truncated

Model Input Representation

Each event is converted into a feature vector.

Token Embedding

- Each token is mapped to a 32-dimensional embedding vector

$$Embedding(token) \rightarrow 32\ features$$

Temporal Feature

- Time gap between events used as a numeric feature

$$time_gap \rightarrow 1\ feature$$

Final Event Feature Vector

$$[Embedding\ (32),\ Time\ Gap\ (1)]$$

Total dimension per event: **33 features**

Sequence Input Shape

$$(batch_size,\ sequence_length,\ features) = (128,\ 150,\ 33)$$

BiLSTM Model Architecture

Core Benefits (Why BiLSTM?)

- Captures temporal behavioral patterns in student activity.
- Utilizes both past and future context in the sequence.
- Effective for modeling sequential student interactions.

Architecture Flow

Token IDs ↓

Embedding Layer (32-dim) ↓

Concatenate Time Gap ↓

BiLSTM Layer (64 hidden units per direction) ↓

Forward State (64) || Backward State (64)

⇓

Concatenated Vector (128-dim) ↓

Fully Connected Layer ↓

Sigmoid ↓

Dropout Probability

Model Training

Data Composition

Each training sample consists of:

$$(\text{token sequence},\ \text{time gap sequence}) \rightarrow \text{dropout label}$$

Training Setup

- **Loss Function:** Binary Cross Entropy
- **Optimizer:** Adam Optimizer
- **Learning Rate:** 0.001
- **Batch Size:** 128 students

Training Objective

Learn behavioral patterns that distinguish active learners from students likely to drop out.

Model Output

Prediction Objective

The model predicts the probability of student dropout:

$$P(\text{dropout} \mid \text{student behavioral sequence})$$

Example Predictions

Student	Dropout Probability
1001	0.87
1002	0.21
1003	0.63

Interpretation

- Higher probability $\rightarrow$ higher dropout risk
- Enables early intervention strategies

Example Behavioral Pattern

$$\textit{video} \rightarrow \textit{video} \rightarrow \textit{problem} \rightarrow \textit{inactivity} \rightarrow \textit{dropout}$$

The BiLSTM learns such behavioral signatures of disengagement.

BiLSTM Dropout Prediction Pipeline

bilstm_pipeline.png

Model Evaluation Results

Performance Metrics

- Accuracy: 0.8587
- Precision: 0.8904
- Recall: 0.9369
- F1 Score: 0.9131
- AUC (ROC): 0.8516

Key Observations

- High recall (93.7%) indicates the model successfully identifies most students who drop out.
- Strong F1-score (0.913) shows a good balance between precision and recall.
- AUC of 0.85 indicates good discrimination between dropout and non-dropout students.

Decile Lift Analysis

- Top decile dropout rate: 96.18%
- Top decile lift: 1.21

Drawbacks of the Current Implementation

Fixed Sequence Length (150)

- Dataset contains highly variable sequences (≈ 17 to > 7000 events)
- Current model truncates long sequences and pads short sequences
- Truncation may remove later interactions which are strong dropout indicators
- Padding also introduces artificial events, which may reduce the efficiency of the LSTM.

Class Imbalance

- Around 80% of students drop out
- Current implementation does not yet apply imbalance correction
- Planned solution: random oversampling or similar balancing methods

Limited Feature Representation

- Current tokens use only source, event type, and category
- Other signals like session structure, activity intensity, and course progress are not included

Limited Model Interpretability

- BiLSTM predictions are difficult to interpret
- Hard to identify which behavioral events contribute most to dropout risk

Future Improvements

- **Addressing Current Drawbacks**
  Handle variable sequence lengths and resolve class imbalance in the dataset.
- **Improved Behavioral Feature Representation**
  Incorporate additional features such as session-level activity patterns, course progression indicators, and learning intensity signals.
- **Advanced Sequence Models**
  Explore architectures such as:
  - Attention-based LSTM models
  - Transformer-based sequence models

  These models can better capture long-range dependencies in student behavior.
- **Model Interpretability**
  Apply interpretability techniques to understand which behavioral patterns contribute most to dropout predictions.
- **Learning Style Integration**
  Incorporate results from a learning style detector and analyze whether